

# Finding a Research Question, Part 1



# ANNOUNCEMENT

- The prompt for your semester project is online. Please familiarize yourself with the prompt. We'll discuss your questions in the next class.
  - **Deliverable 1**: March 22nd, 11:59 pm
  - **Deliverable 2**: April 5th, 11:59 pm
  - **Deliverable 3**: April 19th, 11:59 pm
  - **Deliverable 4**: May 1st, 11:59 pm

# FINDING A RESEARCH QUESTION



Reflect & Discuss:

1. We have learned about paradigms before. Creswell chooses to use the term 'worldview'. What worldview is most closely associated with the scientific method?

# FINDING A RESEARCH QUESTION

- The **postpositivist worldview** is sometimes called the *scientific method* or doing *science research*.
- The major elements of the **postpositivist worldview** are:
  - *Determination* (i.e., causes determine effects or outcomes)
  - *Reductionism* (i.e., reduces the ideas into a small, discrete set to test, such as the variables that comprise hypotheses and research questions)
  - *Empirical observation and measurement*
  - *Theory verification* (i.e., theories need to be tested, verified, and refined).

# FINDING A RESEARCH QUESTION



Reflect & Discuss:

2. What are the philosophical assumptions of the other worldviews?

# FINDING A RESEARCH QUESTION

- The *constructivist worldview* is a philosophical position used in qualitative research.
- The major elements of the *constructivist worldview* are:
  - *Understanding* (i.e., the belief that individuals seek understanding of the world in which they live and work)
  - *Multiple participant meanings* (i.e., individuals develop subjective meanings of their experiences, which are varied and multiple)
  - *Social and historical construction* (i.e., the meanings are formed through historical and social norms)
  - *Theory generation* (i.e., rather than starting with a theory, as in postpositivism, inquirers generate or inductively develop a theory or pattern of meaning)

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- The *transformative worldview* holds that research inquiry needs to be intertwined with politics and a political change agenda to confront social oppression at whatever levels it occurs.
- The major elements of the *transformative worldview* are:
  - *Political*
  - *Power and justice oriented*
  - *Collaborative*
  - *Change oriented*

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- The *pragmatic worldview* is concerned with applications – what works – and solutions to problems.
- The major elements of the *pragmatic worldview* are:
  - *Consequences of actions*
  - *Problem-centered*
  - *Pluralistic* (i.e., multiple methods, different worldviews, assumptions, and forms of data collection and analysis)
  - *Real-world practice-oriented*



# FINDING A RESEARCH QUESTION



Reflect & Discuss:

3. What research methodology/approach is most closely associated with the postpositivist worldview? Describe it.

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- *Quantitative Research:*

- is an approach for *testing objective theories* by *examining the relationship among variables* or *a comparison among groups*.
- These variables, in turn, can be *measured*, typically *on instruments*, so that numbered *data can be analyzed* using *statistical procedures*.
- The final written report has a set structure comprising an introduction, methods, results, and discussion.
- Theories are tested *deductively*, built into a study to protect *against bias, control for alternative or counterfactual explanations*, and seek to *generalize and replicate the findings*.

# FINDING A RESEARCH QUESTION



Reflect & Discuss:

4. Describe some types of quantitative research designs.

# FINDING A RESEARCH QUESTION

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## Quantitative Methods

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Predetermined

Instrument-based questions

Performance data, attitude data,  
observational data, and census  
data

Statistical analysis

Statistical interpretation

- **Survey research:**
  - provides a quantitative or numeric description of a population's trends, attitudes, or opinions by studying a sample of that population. It includes questionnaires or structured interviews for data collection to generalize from a sample to a population.
- **Experimental research:**
  - seeks to determine if a specific treatment influences an outcome. The researcher assesses this by providing a specific treatment to one group, withholding it from another, and then determining how both groups scored on an outcome. It includes true experiments, the random assignment of subjects to treatment conditions, and quasi-experiments that use nonrandomized assignments.



## Reflect & Discuss:

5. Compare examples 1 and 2 in §1.5 Creswell and Creswell, Research design.

***Example 1. Quantitative Approach: Postpositivist Worldview, Experimental Design, and Pretest and Posttest Measures of Attitudes***

- In this scenario, the researcher tests a theory by specifying narrow hypotheses and the collection of data to support or refute the hypotheses. The researcher uses an experimental design to assess attitudes both before and after an experimental treatment. The data are collected on an instrument that measures attitudes, and the information is analyzed using statistical data analysis procedures and hypothesis testing.

***Example 2. Qualitative Approach: Constructivist Worldview, Ethnographic Design, and Observation of Behavior***

- In this situation, the researcher seeks to establish the meaning of a phenomenon from participants' views. This means identifying a culture-sharing group and studying how it develops shared patterns of behavior over time (i.e., ethnography). One of the key elements of collecting data in this way is to observe participants' behaviors during their engagement in activities.

# FINDING A RESEARCH QUESTION



Reflect & Discuss:

6. How are quantitative research methods different from qualitative methods?

# FINDING A RESEARCH QUESTION

| Tends to or Typically                | Qualitative Approaches                                                                                                                                                                                                                                                                                                                                                                       | Quantitative Approaches                                                                                                                                                                                                                                                                  |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Uses these philosophical assumptions | Constructivist/transformational knowledge claims                                                                                                                                                                                                                                                                                                                                             | Postpositivist knowledge claims                                                                                                                                                                                                                                                          |
| Employs these designs                | Phenomenology, grounded theory, ethnography, case study, and narrative                                                                                                                                                                                                                                                                                                                       | Surveys and experiments (true, quasi-experimental, single-subject)                                                                                                                                                                                                                       |
| Employs these methods                | Open-ended questions, emerging approaches, text or image data                                                                                                                                                                                                                                                                                                                                | Closed-ended questions, predetermined approaches, numeric data (may include some open-ended questions)                                                                                                                                                                                   |
| Uses these practices of research     | Positions themselves<br>Collects participant meanings<br>Focuses on a single concept or phenomenon<br>Brings personal values into the study<br>Studies the context or setting of participants<br>Validates the accuracy of findings<br>Makes interpretations of the data<br>Creates an agenda for change or reform<br>Collaborates with the participants<br>Employs text analysis procedures | Tests or verifies theories or explanations<br>Identifies variables to study<br>Relates variables in questions or hypotheses<br>Uses standards of validity and reliability<br>Observes and measures information numerically<br>Uses unbiased approaches<br>Employs statistical procedures |

# FINDING A RESEARCH QUESTION



Reflect & Discuss:

7. Under what circumstances is a quantitative approach favorable?



# FINDING A RESEARCH QUESTION

If the problem requires for

- (a) the identification of factors that influence an outcome,
- (b) the utility of an intervention, or
- (c) understanding the best predictors of outcomes,

then a qualitative approach is the best.

- It is also the best approach to use to test a theory or explanation.

# FINDING A RESEARCH QUESTION



Reflect & Discuss:

8. What is a research question? What are characteristics of good research questions?

# FINDING A RESEARCH QUESTION

- A research question is what a study aims to answer after data analysis and interpretation.
- *Examples:*
  - What are the impacts of recreational fishing on marine ecosystems?
  - How can machine learning-based energy management optimize power distribution in electric vehicles?
  - Why is silicon predominantly used in semiconductor manufacturing?
  - How can emerging smart grid technologies improve the efficiency and reliability of electrical distribution systems?
  - How do vaccination rates and outcomes differ between countries with public vs private healthcare systems?

- A **good research question**:
  - is **feasible** and *interesting* to both the researcher and the scientific community
  - is *manageable*
  - is *novel*
  - is *relevant* (in terms of clarity of the resulting data, specificity of the methodology, objectivity of the outcome, depth of the research, and impact of the study)
  - are *focused, specific, and concise*
  - is *systematic*
  - facilitates understanding of the research topic, objective, scope, and the limitations of the study
  - requires a comprehensive literature search and an in-depth understanding of the problem being investigated
  - is *ethical*
  - has a potential value/publishable
  - ...

# FINDING A RESEARCH QUESTION



Reflect & Discuss:

9. What is a research hypothesis? What are characteristics of a good research hypothesis?

# FINDING A RESEARCH QUESTION

- A *hypothesis* is an educated statement of an expected outcome.
  - This statement is based on background research and current knowledge.
- A *hypothesis* makes a specific prediction about a new phenomenon or a formal statement on the expected relationship between an independent variable and a dependent variable.
- It provides a tentative answer to the research question to be tested or explored.

# FINDING A RESEARCH QUESTION

- Well-constructed **hypotheses** are based on previous reports and verify the research context.
  - These are realistic, in-depth, sufficiently complex, and reproducible.
- More importantly, these **hypotheses** can be addressed and *tested*.

# FINDING A RESEARCH QUESTION

- In short, good hypotheses are
  1. empirically testable;
  2. backed by preliminary evidence;
  3. testable by ethical research;
  4. based on original ideas;
  5. have evidence-based logical reasoning;
  6. can be predicted.



# FINDING A RESEARCH QUESTION



Reflect & Discuss:

10. Identify and describe the types of (research) questions in quantitative research.

# FINDING A RESEARCH QUESTION

- In quantitative research, research questions inquire about the *relationships among variables* being investigated and are usually framed at the start of the study.
- These are *precise* and typically *linked to the subject population, dependent and independent variables, and research design*.

# FINDING A RESEARCH QUESTION

- Research questions may attempt to describe the behavior of a population in relation to one or more variables, or describe the characteristics of variables to be measured (*descriptive research questions*).
- These questions may also aim to discover differences between groups within the context of an outcome variable (*comparative research question*).
- Or, it may elucidate trends and interactions among variables (*relationship research questions*).

# FINDING A RESEARCH QUESTION

## Quantitative research questions

### Descriptive research question

- Measures responses of subjects to variables
- Presents variables to measure, analyze, or assess

What is the proportion of resident doctors in the hospital who have mastered ultrasonography (response of subjects to a variable) as a diagnostic technique in their clinical training?

### Comparative research question

- Clarifies difference between one group with outcome variable and another group without outcome variable

Is there a difference in the reduction of lung metastasis in osteosarcoma patients who received the vitamin D adjunctive therapy (group with outcome variable) compared with osteosarcoma patients who did not receive the vitamin D adjunctive therapy (group without outcome variable)?

- Compares the effects of variables

How does the vitamin D analogue 22-Oxacalcitriol (variable 1) mimic the antiproliferative activity of 1,25-Dihydroxyvitamin D (variable 2) in osteosarcoma cells?

### Relationship research question

- Defines trends, association, relationships, or interactions between dependent variable and independent variable

Is there a relationship between the number of medical student suicide (dependent variable) and the level of medical student stress (independent variable) in Japan during the first wave of the COVID-19 pandemic?

# FINDING A RESEARCH QUESTION



Reflect & Discuss:

11. Identify and describe the types of hypotheses in quantitative research.

**Table 3.** Definitions and examples of quantitative research hypotheses

| Quantitative research hypotheses                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Simple hypothesis</p> <ul style="list-style-type: none"><li>- Predicts relationship between single dependent variable and single independent variable</li></ul> <p>If the dose of the new medication (single independent variable) is high, blood pressure (single dependent variable) is lowered.</p>                                                                                  |
| <p>Complex hypothesis</p> <ul style="list-style-type: none"><li>- Foretells relationship between two or more independent and dependent variables</li></ul> <p>The higher the use of anticancer drugs, radiation therapy, and adjunctive agents (3 independent variables), the higher would be the survival rate (1 dependent variable).</p>                                                |
| <p>Directional hypothesis</p> <ul style="list-style-type: none"><li>- Identifies study direction based on theory towards particular outcome to clarify relationship between variables</li></ul> <p>Privately funded research projects will have a larger international scope (study direction) than publicly funded research projects.</p>                                                 |
| <p>Non-directional hypothesis</p> <ul style="list-style-type: none"><li>- Nature of relationship between two variables or exact study direction is not identified</li><li>- Does not involve a theory</li></ul> <p>Women and men are different in terms of helpfulness. (Exact study direction is not identified)</p>                                                                      |
| <p>Associative hypothesis</p> <ul style="list-style-type: none"><li>- Describes variable interdependency</li><li>- Change in one variable causes change in another variable</li></ul> <p>A larger number of people vaccinated against COVID-19 in the region (change in independent variable) will reduce the region's incidence of COVID-19 infection (change in dependent variable).</p> |

### Causal hypothesis

- An effect on dependent variable is predicted from manipulation of independent variable

A change into a high-fiber diet (independent variable) will reduce the blood sugar level (dependent variable) of the patient.

### Null hypothesis

- A negative statement indicating no relationship or difference between 2 variables

There is no significant difference in the severity of pulmonary metastases between the new drug (variable 1) and the current drug (variable 2).

### Alternative hypothesis

- Following a null hypothesis, an alternative hypothesis predicts a relationship between 2 study variables

The new drug (variable 1) is better on average in reducing the level of pain from pulmonary metastasis than the current drug (variable 2).

### Working hypothesis

- A hypothesis that is initially accepted for further research to produce a feasible theory

Dairy cows fed with concentrates of different formulations will produce different amounts of milk.

### Statistical hypothesis

- Assumption about the value of population parameter or relationship among several population characteristics
- Validity tested by a statistical experiment or analysis

The mean recovery rate from COVID-19 infection (value of population parameter) is not significantly different between population 1 and population 2.

There is a positive correlation between the level of stress at the workplace and the number of suicides (population characteristics) among working people in Japan.

### Logical hypothesis

- Offers or proposes an explanation with limited or no extensive evidence

If healthcare workers provide more educational programs about contraception methods, the number of adolescent pregnancies will be less.

### Hypothesis-testing (Quantitative hypothesis-testing research)

- Quantitative research uses deductive reasoning.

- This involves the formation of a hypothesis, collection of data in the investigation of the problem, analysis and use of the data from the investigation, and drawing of conclusions to validate or nullify the hypotheses.



# FINDING A RESEARCH QUESTION



Reflect & Discuss:

12. What are the steps involved in constructing research questions and hypotheses?



# FINDING A RESEARCH QUESTION

- To construct *effective research questions* and *hypotheses*, it is very important to
  1. clarify the background;
  2. identify the research problem;
  3. review or conduct preliminary research;
  4. construct research questions;
  5. construct specific deductive or inductive predictions in the form of hypotheses;
  6. state the study aims.